



Euler Laboratory, ISAE-Supméca

In cooperation with N. K. Orchid College, Solapur, India



Internship – Mechatronics Modelling of an Entire Industry 5.0 Production Line

Academic level: Master's level (M1-M2)

Duration: 4 months

Period: February 2026 to June 2026

Location: ISAE-Supméca Paris, France

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Internship benefits:

- Eligible for affordable on-campus accommodation
- Access to research facilities and laboratories, seminars and workshops

Compensation: €670/month [+ GIFAS Grant €1300 per student]

Description

This internship aims to develop a unified **process, mechatronic and multiphysics model** of an entire *Industry 5.0* production line, bridging the gap between physical processes, control logic, and intelligent decision-making. The project will contribute to the creation of a **digital twin** that can support smart reconfiguration and optimisation through reinforcement learning (RL) and advanced system identification techniques.

The intern will begin by analysing the production line's architecture and its main mechatronic subsystems (actuators, conveyors, sensors, robots, autonomous ground vehicles, etc.). For each subsystem, a **multiphysics model** will be proposed using Modelica language, covering mechanical, electrical, and thermal behaviours, along with a simplified spatial representation to locate components and potential coupling effects such as vibration transmission, temperature gradients, or other kinds of interference. Each model will also integrate the **production sequence** it contributes to, linking physical dynamics with discrete-event process logic (e.g., activation timing and task dependencies).

A central objective of this internship is to establish a consistent modelling framework where **Modelica-based components are exported as Functional Mock-up Units (FMUs)** and orchestrated from **Python** through the **Functional Mock-up Interface (FMI)** standard. This approach allows the intern to combine high-fidelity physical simulation with Python-based control, optimisation, and data analysis environments. Ultimately, the entire production line should be simulated end-to-end from a single program, enabling real-time integration with learning algorithms or digital twin platforms.

This project will also involve **parameter identification and experimental validation**. The intern will design and execute a series of **minimal-instrumentation experiments** to estimate physical and process parameters and validate the white-box models' predictive capability. This task includes developing a design of experiments (DoE) methodology that ensures robustness and observability while remaining lightweight and feasible on an industrial testbed.

This internship offers a rich combination of **multiphysics, mechatronics and process modelling and simulation for digital twin research**.

Deliverables:

- Structured library of multiphysics and process models
- Technical documentation (report) with identified parameters and coupling analysis
- Python scripts simulating the entire process of the ALIX Industry 5.0 Production Line

Profile

- Background in **Mechanical, Mechatronics, Robotics, or Systems Engineering**.
- Skills in **programming, multiphysics modelling and dynamical systems**.
- Familiarity with **Python, sensors/actuators**. Familiarity with Modelica language is a plus.
- Curiosity and rigor are key to complete this model-based digital twin.